

# Automating Multi-Class Bone Fracture Classification from X-Ray Radiographs

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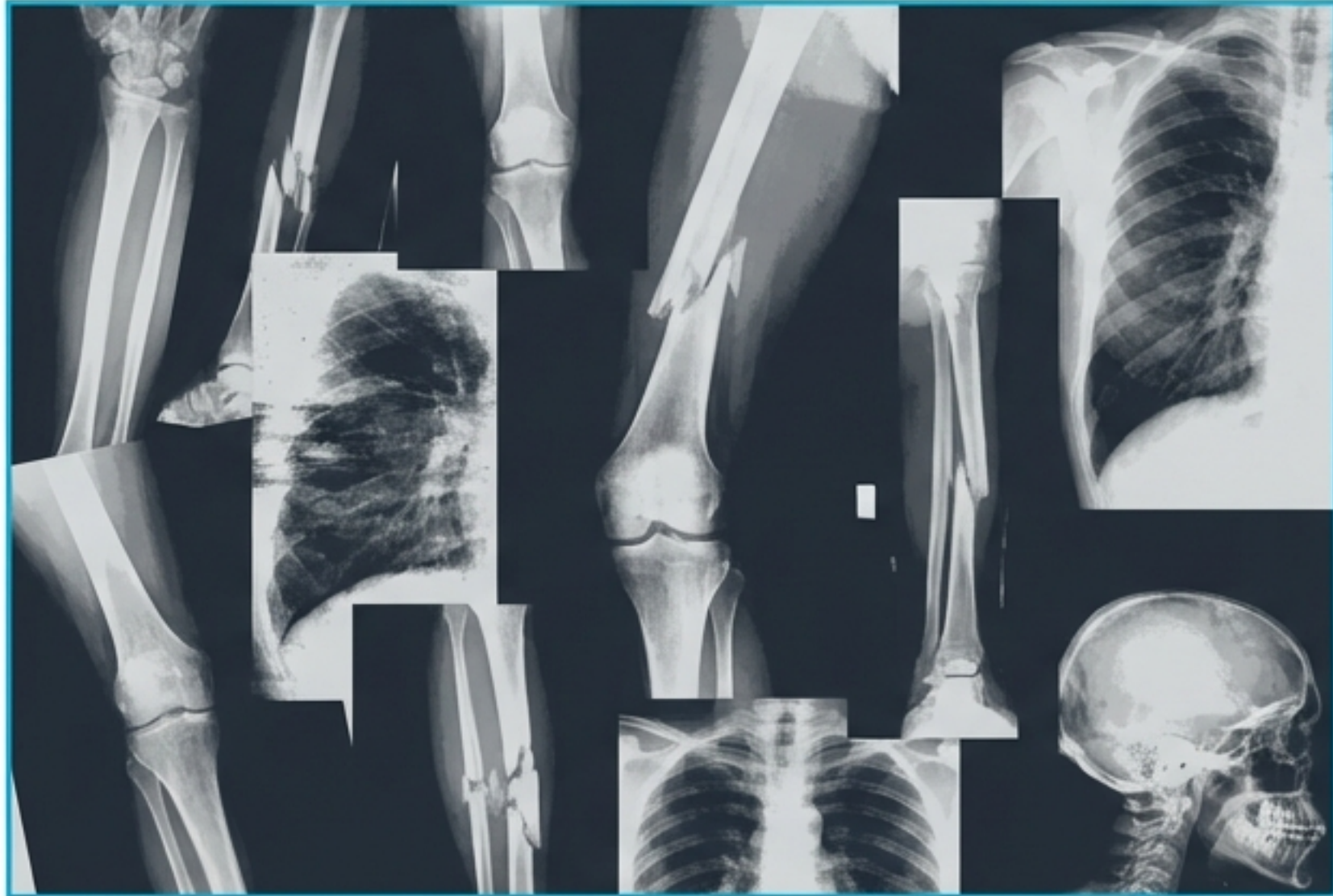
**Problem Statement:** To develop an AI-based clinical decision support tool capable of classifying 17 different types of bone fractures from X-ray radiographs, assisting radiologists in faster, more accurate identification





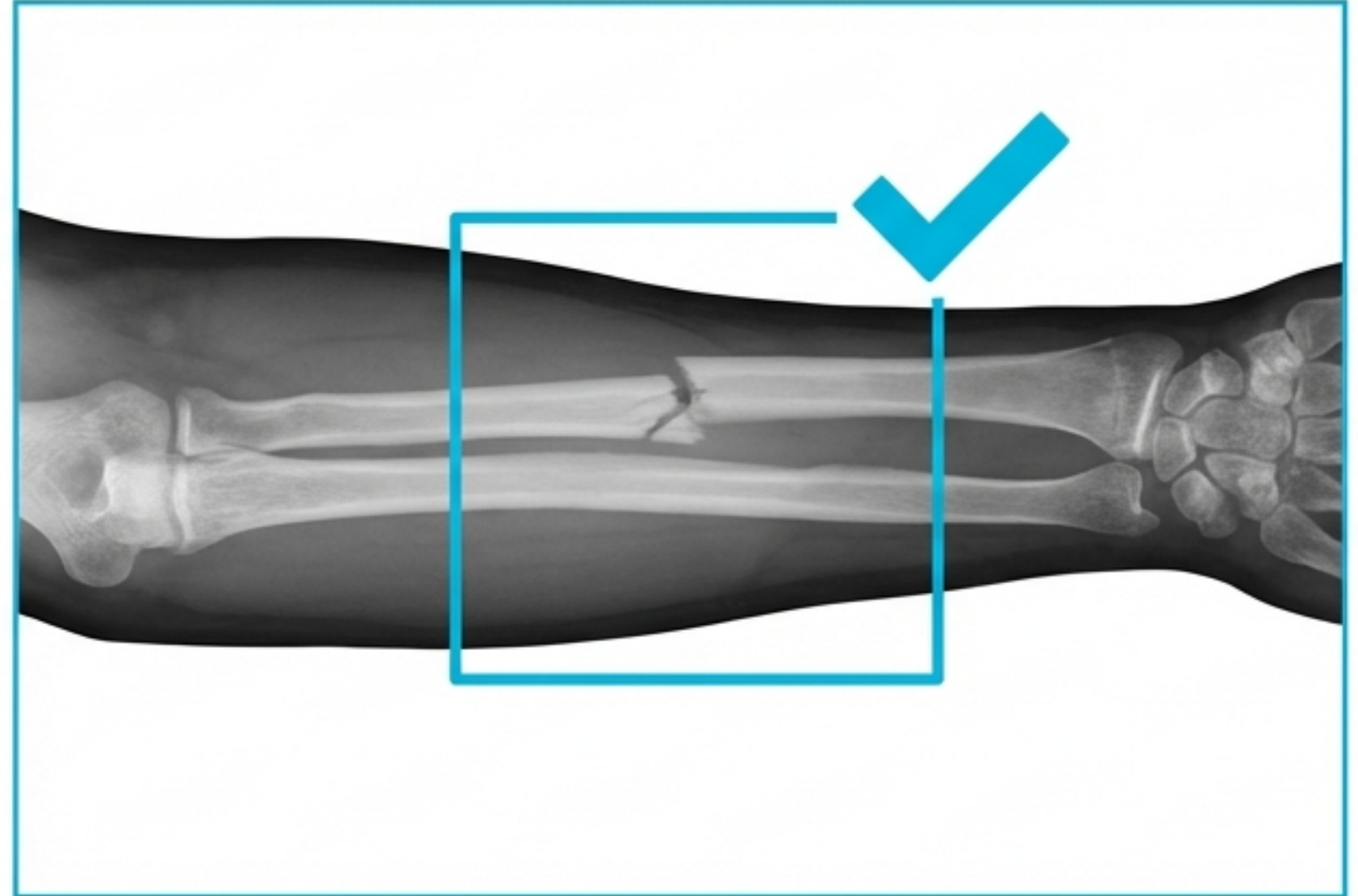
### Clinical Motivation:

Bone fractures are highly common musculoskeletal injuries, but distinguishing between subtle visual subtypes (e.g., simple vs. stress vs. hairline) is complex, time-consuming, and prone to human error



### Project Framing (Objective):

To build a robust AI-based classification system that accurately categorizes complex fracture types, serving as a reliable clinical decision-making support tool for radiologists in emergency settings





# Slide 3: Dataset Overview and Challenges

**Dataset:** Kaggle Multi-Region X-ray Dataset

**Overview:** Contains 4,906–10,580 grayscale PNG/JPEG images spanning 17 distinct fracture classes (e.g., **Comminuted**, Compression-**Crush**, **Greenstick**, Oblique, Spiral, etc.)

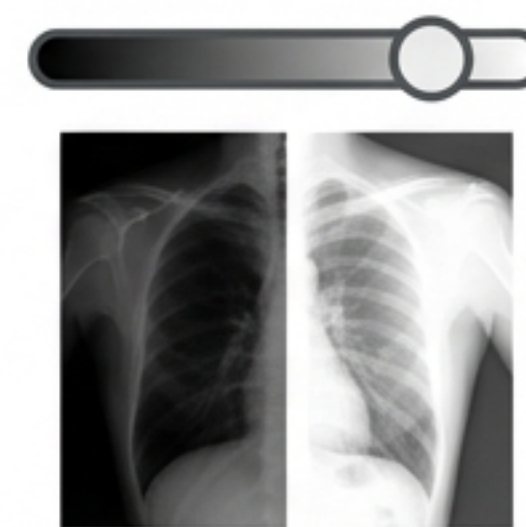
## Challenges Must Overcome:



Extreme class imbalance across the 17 categories



Subtle inter-class visual similarities (e.g., stress fractures masking as normal bone tissue)



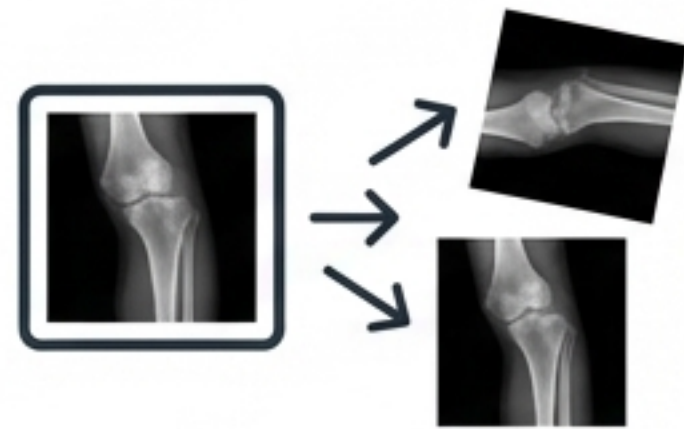
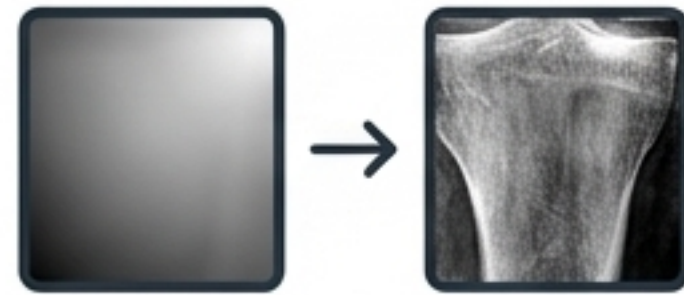
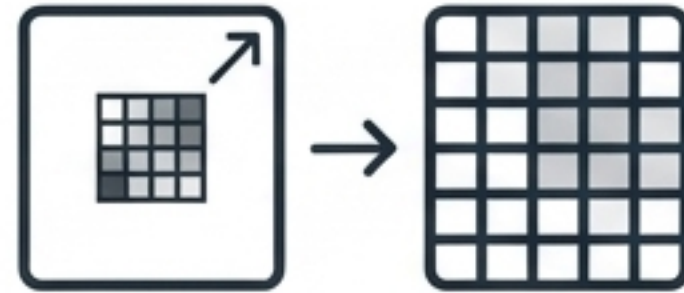
Highly variable baseline radiograph contrast and quality across the dataset



# Slide 4: Preprocessing and Train/Val/Test Strategy

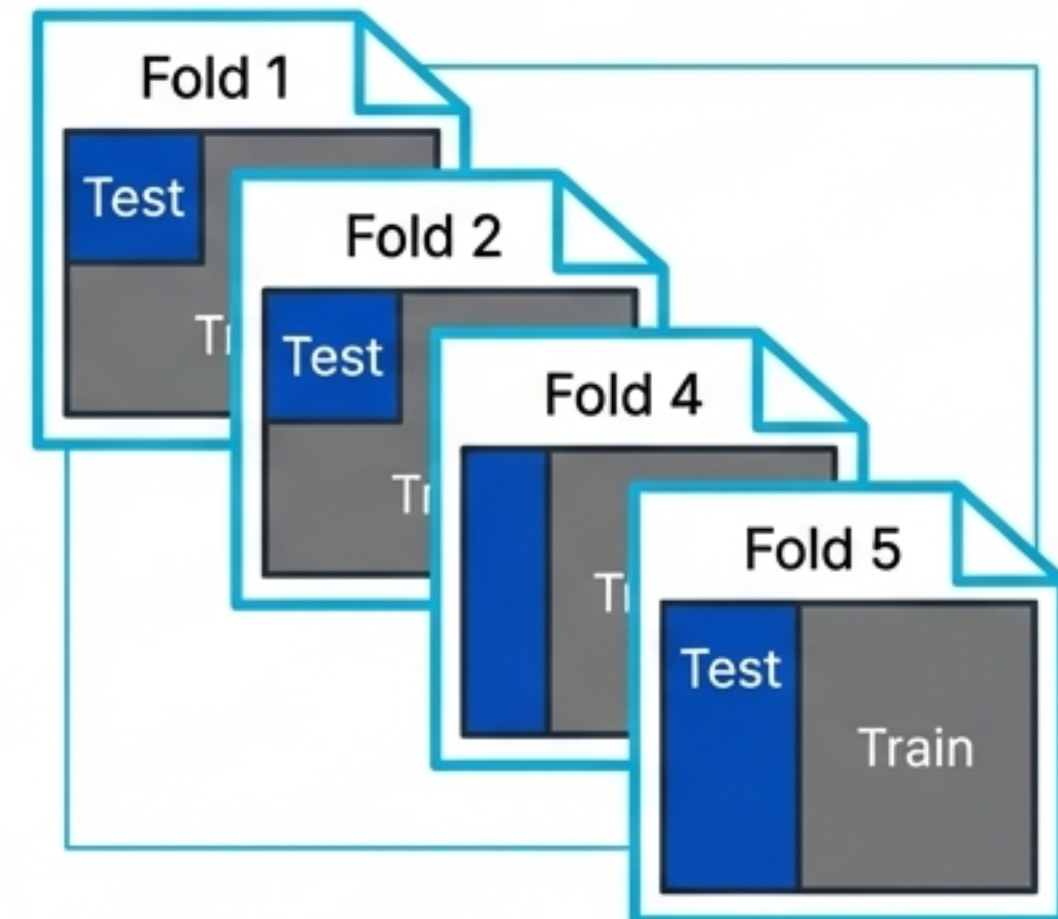
## Preprocessing Pipeline:

- Standardized fixed-resolution resizing (224x224 or 256x256)
- Intensity normalization and Contrast Limited Adaptive Histogram Equalization (CLAHE)
- Data Augmentation: Random rotation, anatomically valid horizontal flipping, and slight zooming to force model generalization



## Split Strategy:

- Strict 70% Train | 10% Validation | 20% Test split
- Stratified splitting utilized to rigorously preserve class ratios
- 5-fold cross-validation employed to ensure robust generalization across variable patient demographics



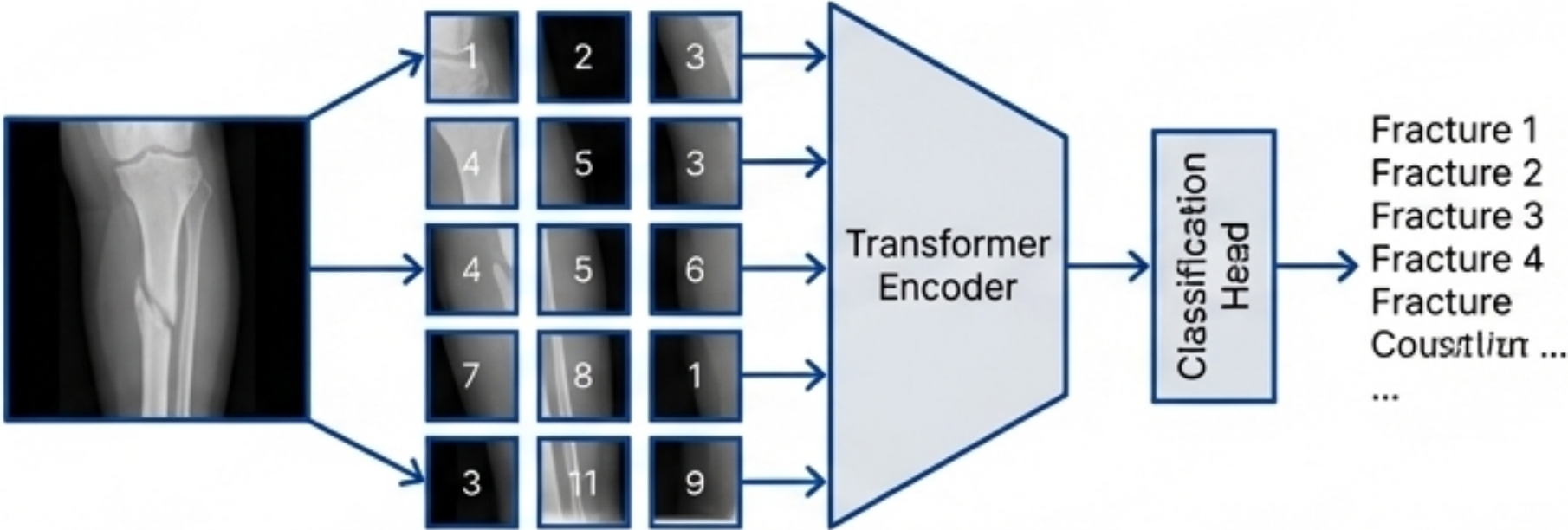


# Slide 5: Model Architecture and Training Strategy

~10.72 Million parameters

## Architecture: Modular Hybrid Vision Transformer (ViT)

**Design Choice:** Leverages transformer attention mechanisms to view radiographs holistically, capturing globally distributed, subtle edge discontinuities better than traditional CNNs



**Rulebook Compliance:** Utilizes approved ImageNet pre-trained weights and strictly avoids prohibited medical-specific pre-trained models

## Training Strategy:

**Loss Function:** Focal Loss or Weighted Cross-Entropy explicitly configured to penalize majority class dominance and focus on hard-to-classify subtle fractures

**Optimization:** AdamW optimizer with a Cosine Annealing learning rate scheduler (starting at 0.0003) for stable convergence

**Early Stopping:** Triggered strictly by Validation Macro F1 to prevent overfitting on the training set



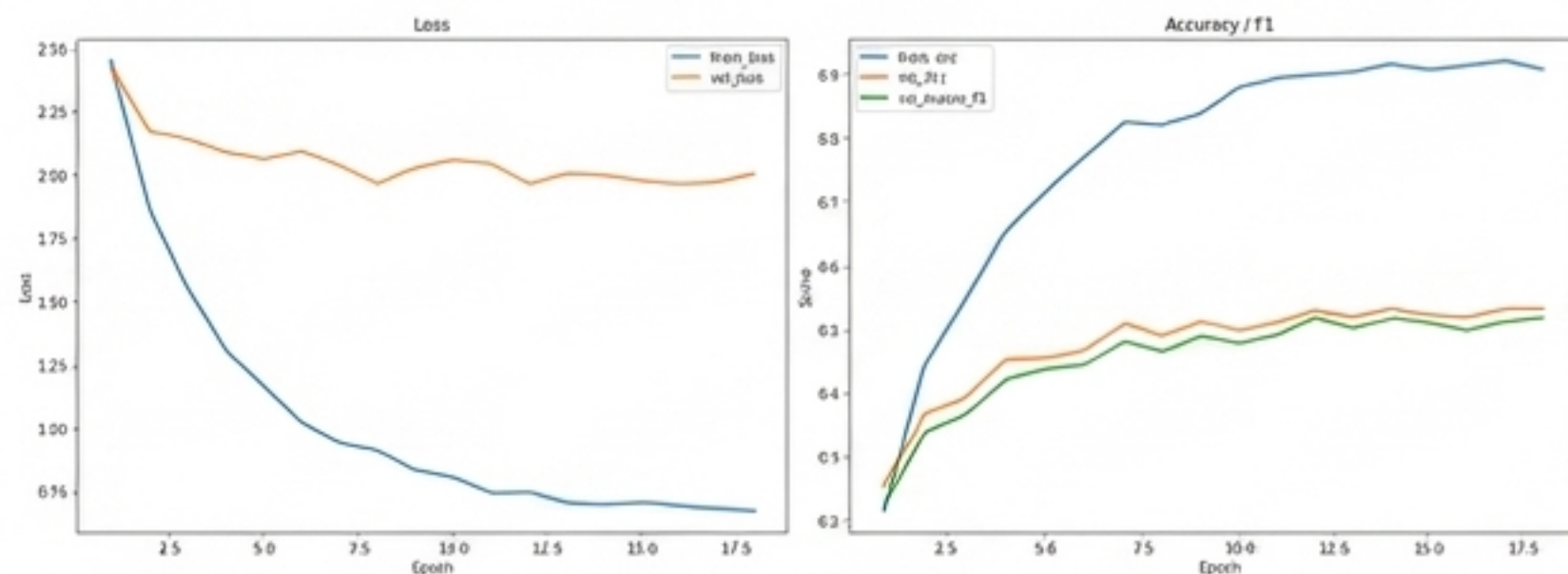
# Slide 6: Results: Curves, Confusion Matrix, ROC

**Overall Test Performance:**  
Accuracy: 51.4% | Top-3 Accuracy: 74.3%

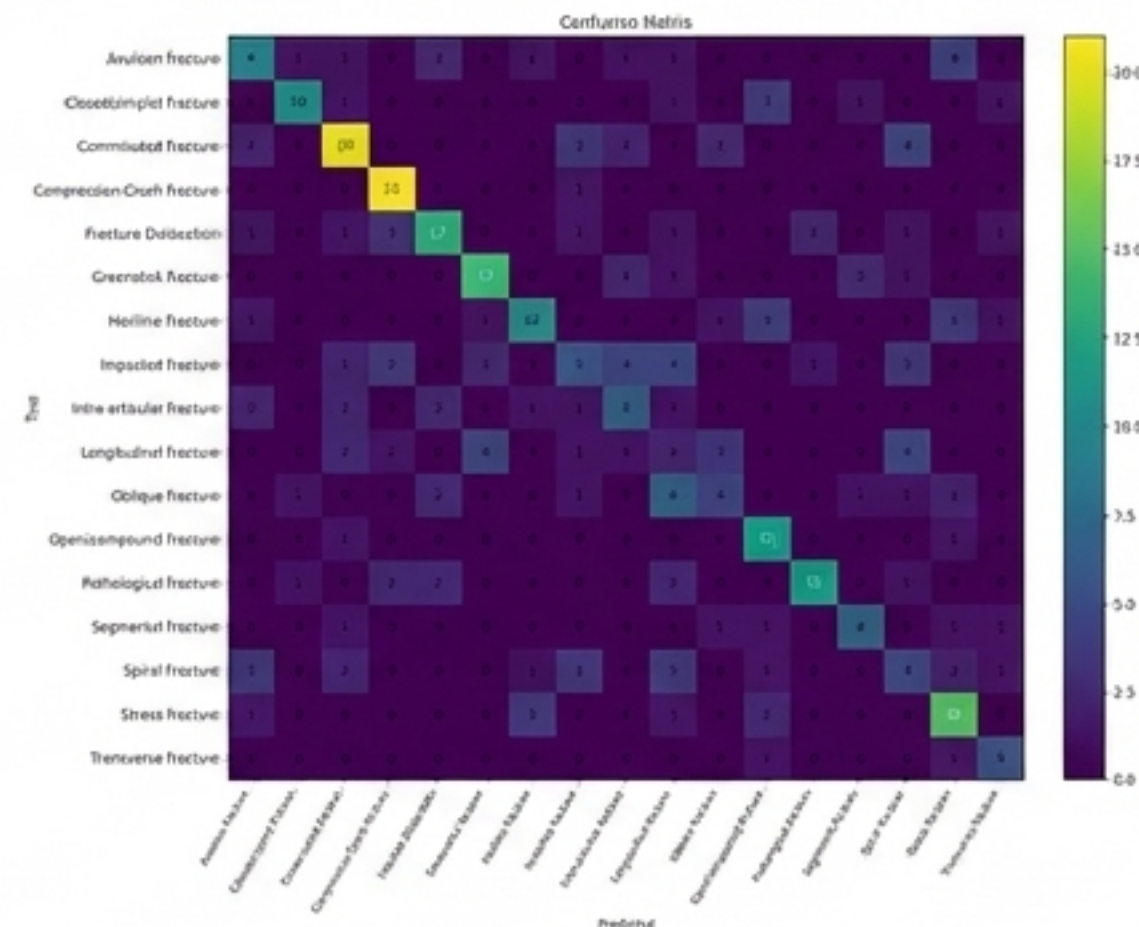
**Macro F1:**  
50.7% | ROC-AUC OVR Macro: 87.4%

**Inference Speed:**  
Highly optimized at ~5.08 ms/image

**Best Performing Classes:** Compression-Crush (F1: 0.81, AUC: 0.99) and Greenstick (F1: 0.68, AUC: 0.97)



**Training Curves:** Validation Macro F1 peaked at ~52.0% at Epoch 12. Training loss steadily decreased to 0.67, though an overfitting gap (~1.33) emerged in later epochs

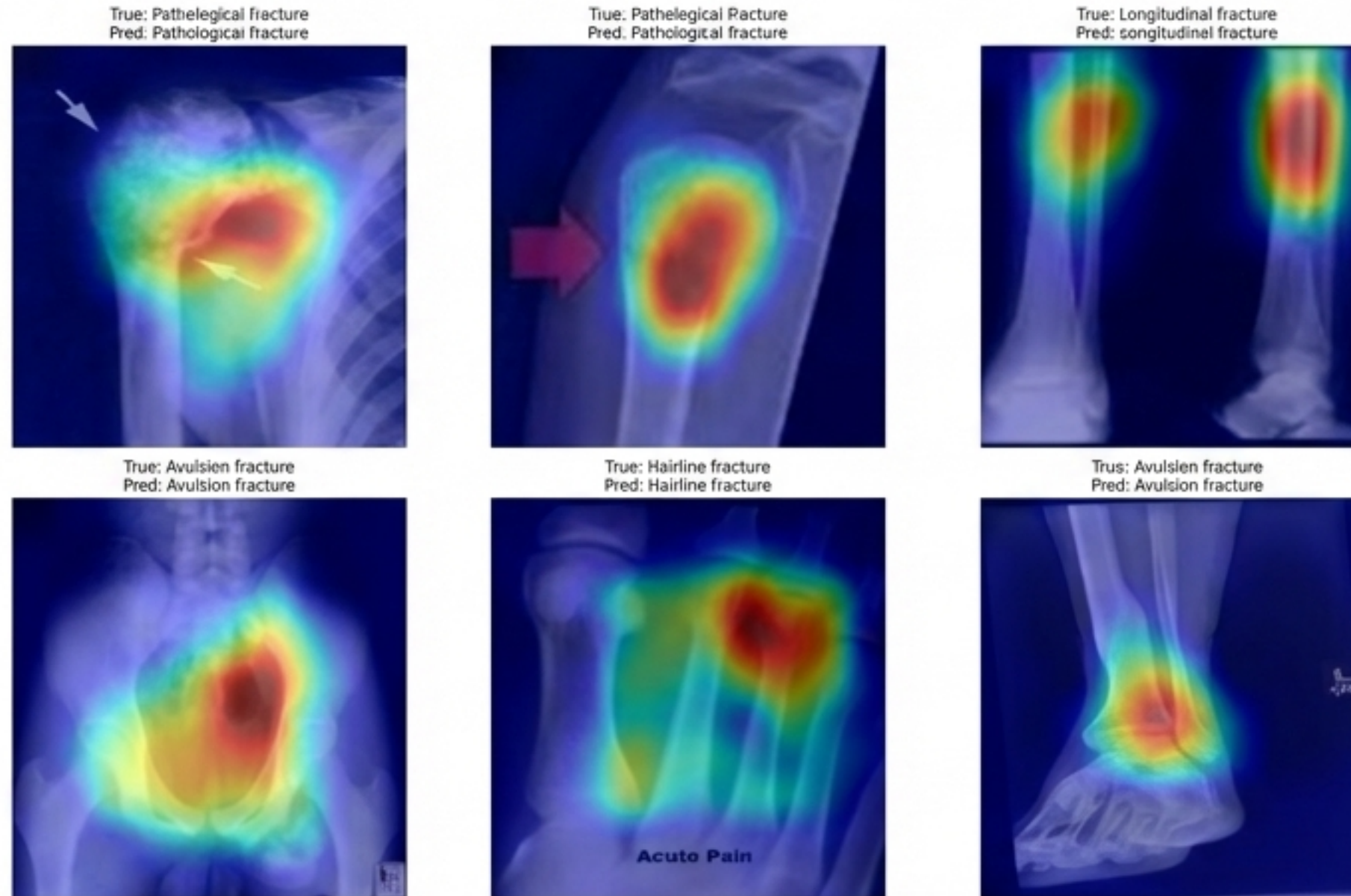


**Confusion Matrix Insights:** The model reliably classifies distinct structural fractures but shows expected confusion between visually adjacent classes (e.g., Longitudinal vs. Oblique)



# Slide 7: Explainability: Grad-CAM and Error Analysis

**Explainability Method:** Attention maps / Grad-CAM overlays



## Transparent Decision Making:

Visualizations build clinical trust by proving the model focuses exclusively on structural bone discontinuities (true fracture lines) rather than relying on a black-box methodology

## Error reon Classes:

Longitudinal fractures (F1: 0.09) highlights the difficulty differentiating subtle linear features from normal bone overlap

## Artifact Rejection:

Attention maps confirm the model correctly ignores background noise, medical hardware implants, and overlaid text artifacts during inference

## Error Analysis:

Low recall in classes like Longitudinal fractures (F1: 0.09) highlights the difficulty of differentiating subtle linear features from normal bone overlap



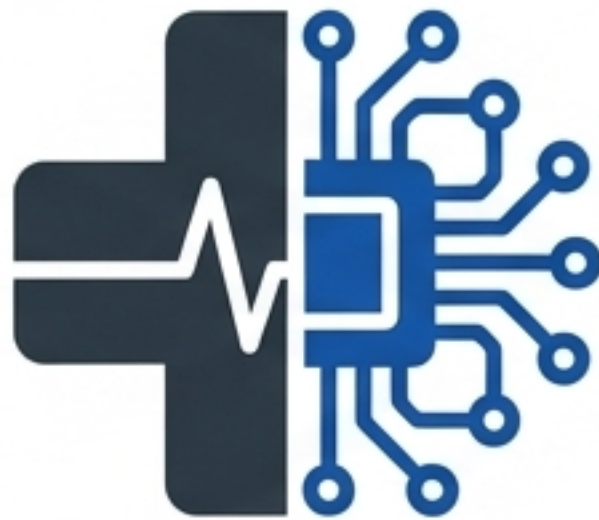
# Slide 8: Challenges, Limitations, Future Work, Final Deliverables

## Limitations:

Model generalization across unseen X-ray hardware and adaptability to varying patient demographic anatomies not fully represented in the training distribution

## Future Improvements:

- Implement an ensemble approach combining CNN local feature extraction with ViT global context
- Optimize model size for low-latency, edge-device deployment directly in hospital emergency rooms
- Expand the dataset to include multi-angle radiograph



## Final Deliverables

- ✓ Finalized Final\_results.csv
- ✓ Model\_performance\_analysis.csv
- ✓ GitHub Codebase
- ✓ Trained Checkpoints submitted